# PRECOMPOSITION ON CONTINUOUS ALTERNATING MAPS IS NOT ANALYTIC IN CHARACTERISTIC 2

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Abstract. For normed spaces $E, E', F$ over a nontrivially normed field $k$ and a finite $\iota$, precomposition $f \mapsto (m \mapsto m \circ (f, \dots, f))$ sends continuous linear maps $E \to E'$ to continuous linear maps between spaces of continuous alternating maps. It is $C^n$ for every $n \in \mathbb{N} \cup \{\infty\}$ in every characteristic [5], and analytic when $|\iota|!$ is invertible in $k$ (by polarisation). We show that it is *not* analytic in general: over every complete ultrametric field of characteristic 2 that is not spherically complete, with $E = E' = \ell^\infty$ and $|\iota| = 2$, analyticity fails at 0. The proof is a finite parity identity for determinants in characteristic 2, glued along $\ell^\infty$; the failure is witnessed by two coordinate projections of $\ell^\infty$. The result is formalized in Lean 4/Mathlib in the form "the general $C^\omega$ statement implies `False`".

## 1. Introduction

Let $k$ be a nontrivially normed field. For normed spaces $E, E', F$ over $k$ and a finite $\iota$ we write $\mathcal{L}(E, E')$ for the space of continuous linear maps with the operator norm, and $\mathrm{Alt}^\iota(E, F)$ for the normed space of continuous alternating $\iota$-linear maps $E^\iota \to F$; we write $\mathrm{Alt}^2$ when $|\iota| = 2$.

**Analytic functors.** A map $f\colon X \to Z$ between normed spaces over $k$ is **analytic at** $x_0 \in X$ if there are bounded $n$-linear maps $p_n\colon X^n \to Z$ and a radius $r > 0$ with $\sup_n \|p_n\| r^n < \infty$ such that $f(x_0 + h) = \sum_{n \geq 0} p_n(h, \dots, h)$ whenever $\|h\| < r$. It is **analytic**, or $C^\omega$, if it is analytic at every point of $X$; this is the definition used in Mathlib [4].

Lang [3, Ch. III, §4] calls a functor $T$ on Banach spaces **of class $C^p$** if $f \mapsto T(f)$ is $C^p$. Similarly, we call a contravariant endofunctor $T$ on the category of normed spaces over $k$ **analytic** if for all $E, E'$ the map

$$\mathcal{L}(E, E') \to \mathcal{L}\big(T(E'), T(E)\big), \qquad f \mapsto T(f),$$

is analytic. For a normed space $F$ and a finite $\iota$, the assignment $E \mapsto \mathrm{Alt}^\iota(E, F)$, $f \mapsto (m \mapsto m \circ (f, \dots, f))$, is a contravariant functor. The assignment is also functorial in $F$: a continuous linear map $g\colon F \to F'$ acts by $m \mapsto g \circ m$. This action is linear in $g$, hence analytic, whereas the action of $f$ is not linear in $f$. So the bifunctor $\mathrm{Alt}^\iota$ is analytic if and only if the functor $\mathrm{Alt}^\iota(-, F)$ is analytic for every $F$.

**Motivation: alternating bundles in Mathlib.** In Mathlib, manifolds and vector bundles are modelled on normed spaces (of possibly infinite dimension) over an arbitrary nontrivially normed field $k$, and their regularity is an element $n \in \mathbb{N} \cup \{\infty, \omega\}$, where $C^\omega$ means analytic. Differential forms require the bundle of continuous alternating maps between two $C^n$ vector bundles to be a $C^n$ vector bundle again (an instance of `ContMDiffVectorBundle`). Its transition functions are those of the given bundles with $\mathrm{Alt}^\iota$ applied, so this comes down to the regularity of the functor $\mathrm{Alt}^\iota(-, F)$, i.e. of the precomposition map

`ContinuousAlternatingMap.compContinuousLinearMapCLM`.

Two cases have been settled, in pull requests to Mathlib.

- If $|\iota|!$ is invertible in $k$ then $\mathrm{Alt}^\iota(-,F)$ is analytic, by polarisation. In Mathlib this is done in characteristic 0 by Gouëzel [1], who uses it to construct the bundle for every $n$ in characteristic 0 [2]; he remarks that a variant of the argument, not included there, covers finite-dimensional spaces in positive characteristic, but not the general case.
- Over every $k$, with no hypothesis on the characteristic or the dimension, $\mathrm{Alt}^\iota(-,F)$ is of class $C^n$ for every $n \in \mathbb{N} \cup \{\infty\}$ [5].

This leaves open the case where $|\iota|!$ is not invertible in $k$ and the spaces are infinite-dimensional: is $\mathrm{Alt}^\iota(-,F)$ analytic there? The following theorem provides a counterexample.

**Theorem 1.** *Let $k$ be a complete ultrametric field of characteristic 2 which is not spherically complete. There is a Banach space $F$ over $k$ such that the functor $\mathrm{Alt}^2(-,F)$ is not analytic.*

Thus the restriction $n \neq \omega$ in [5] cannot be removed. In characteristic $\ell \geq 3$ the case $|\iota| \geq \ell$ remains open (Remark 17). The result is machine-checked in Lean 4 with Mathlib (§9).

## 2. Statement

**Notation.** For a polynomial $f$ and a monomial $\mu$, $\mathrm{coeff}_\mu f$ is the coefficient of $\mu$ in $f$. We write $c_0 \subset \ell^\infty$ for the null sequences, $\mathbf{1}_S$ for the indicator of a set $S$, and $\mathbf{1} = \mathbf{1}_{\mathbb{N}}$.

Throughout, $k$ is a field which is complete for a nontrivial absolute value $|\cdot|$ satisfying the **ultrametric inequality** $|x+y| \leq \max(|x|,|y|)$; we call such a field **ultrametric**. The sup norms used below inherit the inequality. We assume that

(K1) $\mathrm{char}\, k = 2$;
(K2) $k$ is not spherically complete: there is a sequence of closed balls $B(c_n, r_n)$ such that $|c_{n+1} - c_n| \leq r_n$, $r_0 > r_1 > \cdots$ and $\bigcap_n B(c_n, r_n) = \emptyset$.

The balls in (K2) are balls in $k$, and they are nested by the ultrametric inequality. Such fields exist; an explicit one is given in §7. Let $E = \ell^\infty = \ell^\infty(\mathbb{N}, k)$ with the sup norm, a Banach algebra under pointwise multiplication; write $m_a \in \mathcal{L}(E,E)$ for multiplication by $a \in E$, so that $\|m_a\| = \|a\|$, and $e_i$ for the unit vectors. For $x, y \in E$ let

$$W(x,y) := x \otimes y + y \otimes x \in \ell^\infty(\mathbb{N} \times \mathbb{N}, k), \qquad W(x,y)_{pq} = x_p y_q + x_q y_p,$$

and let $F \subseteq \ell^\infty(\mathbb{N} \times \mathbb{N}, k)$ be the closed linear span of all $W(x,y)$, a Banach space. Let

$$Q \colon \mathcal{L}(E,E) \to \mathcal{L}\big(\mathrm{Alt}^2(E,F), \mathrm{Alt}^2(E,F)\big), \qquad Q(f)(m) = m \circ (f,f).$$

**Theorem 2.** *$Q$ is not analytic at 0. In particular $Q$ is not $C^\omega$.*

The map $Q$ is the action on $\mathcal{L}(E,E)$ of the functor $\mathrm{Alt}^2(-,F)$, so Theorem 2 implies Theorem 1. Sections 3–6 prove Theorem 2 in four steps.

## 3. Step 1: analyticity forces a bilinear lift

**Lemma 3.** *Let $Q \colon X \to Z$ be a map between normed spaces with $Q(tf) = t^2 Q(f)$. If $Q$ is analytic at 0 then there is a bounded bilinear $P \colon X \times X \to Z$ with $P(f,f) = Q(f)$ for all $f$.*

*Proof.* Let $\sum_n p_n$ be a power series of $Q$ at 0, and fix $f \in X$. For scalars $t \in k$ with $|t|$ small,

$$t^2 Q(f) = Q(tf) = \sum_n t^n\, p_n(f, \ldots, f).$$

Both sides are power series in the variable $t \in k$ with coefficients in $Z$. The coefficients of a convergent power series in one variable are unique over any nontrivially normed field, so comparing the coefficients of $t^2$ gives $p_2(f,f) = Q(f)$. Take $P = p_2$. □

So assume that a bounded bilinear $P$ with $P(f,f) = Q(f)$ exists. The map $W\colon E \times E \to F$ is bilinear, it is alternating because $W(x,x) = 2\,x \otimes x = 0$, and it is continuous because $\|W(x,y)\| \le \|x\|\|y\|$ by the ultrametric inequality; so $W \in \mathrm{Alt}^2(E,F)$ with $\|W\| \le 1$. Put, for $a, b, x, y \in E$,

$$\Phi(a,b;x,y) := P(m_a, m_b)(W)(x,y) \in F.$$

Then $\Phi$ is 4-linear, and since $\|m_a\| = \|a\|$ and $\|W\| \le 1$ we have

$$\|\Phi(a,b;x,y)\| \le C\|a\|\,\|b\|\,\|x\|\,\|y\|, \qquad C := \|P\|.$$

Here $C \ge 1$, because $P(\mathrm{id},\mathrm{id}) = Q(\mathrm{id})$ is the identity map of $\mathrm{Alt}^2(E,F) \neq 0$. Moreover

$$\Phi(a,b;x,x) = 0, \qquad \Phi(a,a;x,y) = W(ax,ay). \tag{1}$$

Since $2 = 0$, the first identity gives the symmetry $\Phi(a,b;x,y) = \Phi(a,b;y,x)$. We will contradict the existence of such a $\Phi$.

## 4. Step 2: a finite parity identity

Fix a finite set $I \subset \mathbb{N}$ of even size $2m$ and $a', b' \in E$ with $\|a'\|, \|b'\| \le 1$. Let $X_i, Y_i$ $(i \in I)$ be indeterminates. Expanding $\Phi$ multilinearly in $x = \sum_{i\in I} X_i e_i$ and $y = \sum_{i\in I} Y_i e_i$ defines, for any fixed $p, q \in I$, a polynomial

$$G_{pq} := \Phi(a' + x,\, b' + y;\, a' + x,\, b' + y)_{pq} \in k[X_i, Y_i]_{i\in I}$$

in the polynomial ring in these $4m$ variables, which we abbreviate to $k[X,Y]$. The values of the $G_{pq}$ at points $x, y \in k^I$ (vectors supported on $I$) are the entries of an $I \times I$ matrix $G(x,y)$. We also put

$$g_{pq} := \Phi(x,y;x,y)_{pq} = \sum_{i,j,k,l\in I} \Phi(e_i,e_j;e_k,e_l)_{pq}\, X_iY_jX_kY_l \in k[X,Y].$$

**Proposition 4.** *The matrix $G = (G_{pq})_{p,q\in I}$ is symmetric with zero diagonal: $G_{pq} = G_{qp}$ and $G_{pp} = 0$ in $k[X,Y]$.*

*Proof.* Every element of $F$ is a symmetric matrix with zero diagonal, since $W(x,y)_{pq} = W(x,y)_{qp}$ and $W(x,y)_{pp} = 2x_py_p = 0$, and both properties pass to the closed span. The coefficients of $G_{pq}$ are the $(p,q)$ entries of values of $\Phi$, which lie in $F$. □

**Proposition 5.** *The polynomial $g_{pq}$ is homogeneous of degree 2 in the $X_i$ and of degree 2 in the $Y_i$, and $G_{pq} - g_{pq}$ has total degree $\le 3$. In particular $G_{pq}$ has total degree $\le 4$.*

*Proof.* By 4-linearity, $G_{pq}$ is a sum of 16 terms, according to whether each of the four slots is filled by the constant vector ($a'$ or $b'$) or by the variable one ($x$ or $y$), and the total degree of a term is the number of its variable slots. So $g_{pq}$, which has two slots with $x$ and two with $y$, is exactly the degree 4 part of $G_{pq}$, and $G_{pq} - g_{pq}$ has total degree $\le 3$. □

**Proposition 6.** *For $p \neq q$ we have, in $k[X,Y]$,*

$$g_{pq}(Y,X) = g_{pq}(X,Y) + X_p^2Y_q^2 + X_q^2Y_p^2 \qquad \text{and} \qquad g_{pq}(X,X) = 0.$$

*Proof.* The field $k$ is infinite, since it has a nontrivial absolute value, and a polynomial over an infinite field which vanishes at every point is zero. So it suffices to check these identities at

points $x, y \in k^I$. By the symmetry in the last two slots, $g(y,x) = \Phi(y,x;y,x) = \Phi(y,x;x,y)$, so by bilinearity in the first two slots and (1)

$$\begin{aligned} g(x,y) + g(y,x) &= \Phi(x+y, x+y; x, y) + \Phi(x,x;x,y) + \Phi(y,y;x,y) \\ &= W\big((x+y)x, (x+y)y\big) + W(x^2, xy) + W(xy, y^2) \\ &= \big[W(x^2,xy) + W(x^2,y^2) + W(xy,xy) + W(xy,y^2)\big] \\ &\quad + W(x^2,xy) + W(xy,y^2) \\ &= W(x^2, y^2), \end{aligned}$$

where the last step uses $2 = 0$ and $W(xy, xy) = 0$. This is the matrix whose $(p,q)$ entry is $x_p^2 y_q^2 + x_q^2 y_p^2$. Finally $g(x,x) = W(x^2, x^2) = 0$. □

Let $\tau$ be the automorphism of $k[X,Y]$ swapping $X_i \leftrightarrow Y_i$, and $\delta$ the substitution $Y_i := X_i$.

**Lemma 7** (balanced coefficients)**.** *Let $f \in k[X,Y]$ with $\tau f = f$. For every $S \subseteq I$ the coefficient of $\prod_{i\in S} X_i Y_i$ in $f$ equals the coefficient of $\prod_{i\in S} X_i^2$ in $\delta f$.*

*Proof.* The coefficient of $\prod_S X_i^2$ in $\delta f$ is the sum of the coefficients in $f$ of all monomials $X^\alpha Y^\beta$ with $\alpha + \beta = 2 \cdot \mathbf{1}_S$. The involution $(\alpha, \beta) \mapsto (\beta, \alpha)$ of this set of monomials preserves coefficients (as $\tau f = f$), and its only fixed point is $\alpha = \beta = \mathbf{1}_S$; all other terms cancel in pairs. Thus the coefficient of $\prod_S X_i^2$ in $\delta f$ is equal to the coefficient of $X^{\mathbf{1}_S} Y^{\mathbf{1}_S} = \prod_{i\in S} X_i Y_i$ in $f$. □

A **perfect matching** of $I$ is a partition of $I$ into $m$ unordered pairs $\{p,q\}$, its edges; for example $\{1,2,3,4\}$ has the three perfect matchings $12|34$, $13|24$ and $14|23$. Since $G_{pq} = G_{qp}$ (Proposition 4), the product of the $G_{pq}$ over the edges of a perfect matching is well defined.

**Proposition 8.** *Let $M$ be a perfect matching of $I$ and $\nu := \prod_{i\in I} X_i Y_i$. Then*

$$\operatorname{coeff}_\nu \prod_{\{p,q\}\in M} G_{pq} = 1.$$

*Proof.* Write $G_{pq} = g_{pq} + c_{pq}$, where $g_{pq}$ is homogeneous of total degree 4 and $c_{pq}$ has total degree $\le 3$ (Proposition 5). Expanding the product over the $m$ edges of $M$ gives $\prod_M g_{pq}$ plus terms that contain at least one factor $c_{pq}$, and each such term has total degree $\le 3 + 4(m-1) = 4m-1$. The monomial $\nu$ has total degree $4m$, so only $\prod_M g_{pq}$ contributes to its coefficient:

$$\operatorname{coeff}_\nu \prod_M G_{pq} = \operatorname{coeff}_\nu \prod_M g_{pq}.$$

Put $g'_{pq} := g_{pq} + X_p^2 Y_q^2$. By Proposition 6,

$$\tau g'_{pq} = g'_{pq} \qquad \text{and} \qquad \delta g'_{pq} = X_p^2 X_q^2.$$

A product containing a factor $X_p^2 Y_q^2$ has no squarefree monomial, so replacing every $g_{pq}$ by $g'_{pq}$ does not change the coefficient of $\nu$. By Lemma 7, applied to the $\tau$-invariant product $\prod_M g'_{pq}$, this coefficient is the coefficient of $\prod_{i\in I} X_i^2$ in

$$\delta \prod_M g'_{pq} = \prod_{\{p,q\}\in M} X_p^2 X_q^2 = \prod_{i\in I} X_i^2,$$

which is exactly 1. □

**Lemma 9.** *Let $A$ be a symmetric matrix with zero diagonal over a commutative ring of characteristic 2, of even size $2m$. Then $\det A = \sum_M \prod_{\{p,q\}\in M} A_{pq}^2$, the sum being over perfect matchings; and the number of perfect matchings is $(2m-1)!!$.*

*Proof.* In characteristic 2 we have $\det A = \sum_\sigma \prod_i A_{i\sigma(i)}$. As $A$ is symmetric, $\sigma$ and $\sigma^{-1}$ give the same product, so the permutations with $\sigma \neq \sigma^{-1}$ cancel in pairs and only the involutions remain. An involution with a fixed point $i$ contributes the factor $A_{ii} = 0$. An involution without fixed points is a product of $m$ disjoint transpositions $(p\ q)$; the pairs $\{p,q\}$ form a perfect matching $M$, every perfect matching arises exactly once in this way, and therefore

$$\det A = \sum_M \prod_{\{p,q\}\in M} A_{pq}A_{qp} = \sum_M \prod_{\{p,q\}\in M} A_{pq}^2.$$

Finally, pairing the smallest index with one of the $2m-1$ others and continuing with the remaining indices shows that there are $(2m-1)(2m-3)\cdots 3\cdot 1 = (2m-1)!!$ perfect matchings. □

**Lemma 10** (hypercube sum)**.** *Let $R$ be a commutative ring of characteristic 2 and let $f \in R[Z_1,\dots,Z_N]$ have total degree $\le N$. Then $\sum_{z\in\{0,1\}^N} f(z)$ is the coefficient of $Z_1Z_2\cdots Z_N$ in $f$.*

*Proof.* By linearity we may take $f$ to be a monomial $Z^\alpha$. At a point $z \in \{0,1\}^N$ it takes the value 1 if $z_i = 1$ for all $i$ in the support of $\alpha$, and 0 otherwise; so the sum is $2^{N-|\operatorname{supp}\alpha|}$, which vanishes in $R$ unless every variable occurs in $Z^\alpha$. The only monomial of degree $\le N$ in which all $N$ variables occur is $Z_1\cdots Z_N$. □

**Corollary 11.** *We have*

$$\sum_{x,y\in\{0,1\}^I} \det G(x,y) = 1,$$

*and consequently there are $x, y \in \{0,1\}^I$ with $|\det G(x,y)| \ge 1$.*

*Proof.* Each $G_{pq}$ has total degree $\le 4$ (Proposition 5), so for a perfect matching $M$ the product $\prod_{\{p,q\}\in M} G_{pq}$ has total degree $\le 4m$, which is the number of variables $X_i, Y_i$. By Lemma 10 its sum over all $x, y \in \{0,1\}^I$ is therefore the coefficient of the monomial $\prod_{i\in I} X_iY_i$, in which every variable occurs once, and this coefficient is 1 by Proposition 8. For each pair $x, y$ the matrix $G(x,y)$ is symmetric with zero diagonal (Proposition 4), so Lemma 9 applies to it. In characteristic 2 we have $(a+b)^2 = a^2+b^2$, so a sum of squares is the square of the sum. Hence

$$\begin{aligned}\sum_{x,y\in\{0,1\}^I} \det G(x,y) &= \sum_{x,y\in\{0,1\}^I} \sum_M \prod_{\{p,q\}\in M} G_{pq}(x,y)^2 \\ &= \sum_M \sum_{x,y\in\{0,1\}^I} \Big( \prod_{\{p,q\}\in M} G_{pq}(x,y)\Big)^2 \\ &= \sum_M \Big( \sum_{x,y\in\{0,1\}^I} \prod_{\{p,q\}\in M} G_{pq}(x,y)\Big)^2 = \sum_M 1^2 = (2m-1)!! = 1,\end{aligned}$$

because $(2m-1)!!$ is a product of odd numbers, hence odd. If all the values $\det G(x,y)$ had absolute value $< 1$, the ultrametric inequality would give $|1| < 1$. □

The point is that the coefficient 1 does not depend on the unknown lift $P$: all of its freedom is carried by $\tau$-invariant terms vanishing on the diagonal $Y = X$, which Lemma 7 cannot see. No single pair $(x, y)$ works for every lift, but the $2^{4m}$ vertices of the cube cannot all fail at once.

## 5. Step 3: locality on $\ell^\infty$

For $x \in \ell^\infty$ write $x|_{\geq N}$ for $x$ with its coordinates below $N$ set to 0.

**Proposition 12.** *Under* (K2)*, a bounded linear functional $\varphi$ on $\ell^\infty$ with $\varphi(e_i) = 0$ for all $i$ is zero.*

*Proof.* $\varphi$ kills finitely supported sequences, so $|\varphi(x)| \leq \|\varphi\| \sup_{i \geq M} |x_i|$ for every $M$. Suppose $\varphi \neq 0$. For every $0 < s < \|\varphi\|$ there is $w$ with $\varphi(w) = 1$ and $\|w\| \leq 1/s$; fix one such vector $w_0$. Because $r_{n+1} < r_n$ we can choose $s_n \in \ell^\infty$ with $\varphi(s_n) = c_{n+2} - c_{n+1}$ and $\|s_n\| \leq r_n/\|\varphi\|$: take $s_n = (c_{n+2} - c_{n+1})w$ with $\|w\| \leq r_n/(r_{n+1}\|\varphi\|)$. Let $x_n := c_1 w_0 + \sum_{k<n} s_k$; then $\varphi(x_n) = c_{n+1}$ and $\|x_{n+d} - x_n\| \leq r_n/\|\varphi\|$ by the ultrametric inequality. The diagonal sequence $y_i := x_i(i)$ is bounded, since $\|x_i\| \leq \max(\|x_0\|, r_0/\|\varphi\|)$, and $|y_i - x_n(i)| \leq r_n/\|\varphi\|$ for $i \geq n$. Hence $|\varphi(y) - c_{n+1}| = |\varphi(y - x_n)| \leq r_n$, and since $|c_{n+1} - c_n| \leq r_n$ the ultrametric inequality gives $|\varphi(y) - c_n| \leq r_n$ for every $n$. So $\varphi(y) \in \bigcap_n B(c_n, r_n)$, a contradiction. □

This is the direction of van der Put's theorem [6] that we need, $(\ell^\infty/c_0)' = 0$; the index shift absorbs the usual $(1 + \varepsilon)$ loss, and no Hahn–Banach argument is used.

**Proposition 13** (multilinear Phillips property)**.** *Let $\lambda$ be a bounded $d$-linear form on $\ell^\infty$ and $\varepsilon > 0$. There is $N$ such that $|\lambda(u_1, \ldots, u_d)| \leq \varepsilon$ whenever $\|u_j\| \leq 1$ for all $j$ and $u_j = u_j|_{\geq N}$ for some $j$.*

*Sketch* (the complete proof is in the Lean development)*. The case $d = 1$, $\lambda = \varphi$.* (a) $\varphi(e_i) \to 0$. Otherwise $|\varphi(e_{i_k})| \geq \varepsilon$ along a subsequence. Let $T$ be the bounded operator with $T(x)_{i_k} = x_k/\varphi(e_{i_k})$ and all other coordinates 0, and $\psi := \varphi \circ T$, so that $\psi(e_k) = 1$ for all $k$. With sh the right shift, $\psi \circ (1 - \mathrm{sh})$ kills every $e_k$, hence vanishes by Proposition 12; but $(1 - \mathrm{sh})\mathbf{1} = e_0$, so $0 = \psi(e_0) = 1$. (b) For fixed $x$, $\varphi(x|_{\geq N}) - \varphi(x|_{\geq N+1}) = x_N \varphi(e_N) \to 0$, so $\varphi(x|_{\geq N})$ converges; the limit is a bounded linear functional of $x$ killing every $e_i$, hence it is 0. (c) If the claim failed for $\varepsilon$, then (b) would give unit vectors $w_k$ with disjoint finite supports moving to infinity and $|\varphi(w_k)| > \varepsilon$; the functional $c \mapsto \varphi(\sum_k c_k w_k)$ (coordinatewise sum) then contradicts (a).

*The case $d > 1$,* by induction on $d$. Suppose the claim fails for $\varepsilon$. First, for every $M$ there is a unit tuple with $|\lambda| > \varepsilon$ *all* of whose slots vanish below $M$: start from a witness one of whose slots is supported very far out, and split any other slot into its part below $M$, a finite combination of the $e_i$, and its tail; the inductive hypothesis, applied to the finitely many $(d-1)$-linear forms obtained by freezing that slot at $e_i$, $i < M$, shows that the part below $M$ contributes at most $\varepsilon$, so the tail carries $|\lambda| > \varepsilon$. Next, by (b) each slot can be cut down to a finite window. Finally choose such tuples $z^{(s)}$ with windows moving to infinity fast enough and let $x$ be the coordinatewise sum of their first slots. Then $\lambda(x, z_2^{(s)}, \ldots, z_d^{(s)})$ equals $\lambda(z^{(s)})$, plus finitely many terms with an earlier first slot (each $\leq \varepsilon$ by the inductive hypothesis for $\lambda(z_1^{(r)}, \cdot)$, $r < s$), plus one term with a far first slot ($\leq \varepsilon$ by the case $d = 1$). So it has absolute value $> \varepsilon$ for every $s$, contradicting the inductive hypothesis for the $(d-1)$-linear form $\lambda(x, \cdot)$. □

This is exactly where $\ell^\infty$ and (K2) are used; the statement is false for $c_0$ and for spherically complete fields.

## 6. Step 4: gluing

Choose inductively finite sets $I_1 < I_2 < \cdots$ in $\mathbb{N}$ (every element of $I_t$ is smaller than every element of $I_{t+1}$) with $|I_t| = 2t$ and vectors $a_t, b_t \in \{0,1\}^{I_t}$ as follows. Given the earlier stages let $a' = \sum_{s<t} a_s$, $b' = \sum_{s<t} b_s$, and apply Corollary 11 to get $a_t, b_t$ with $|\det G_t| \geq 1$, where $G_t = \Phi(a' + a_t, b' + b_t; a' + a_t, b' + b_t)|_{I_t \times I_t}$. Put $\varepsilon_t := \frac{1}{2} C^{-2t}$ and use Proposition 13 (with

$d = 4$) for the finitely many entry forms $\Phi(\cdot)_{pq}$, $p, q \in I_t$, to place all later sets $I_s$, $s > t$, so far out that these forms are at most $\varepsilon_t$ on unit arguments one of which is supported there.

Let $a, b \in \ell^\infty$ be defined by $a|_{I_t} = a_t$, $b|_{I_t} = b_t$ and zero elsewhere. These are not norm-convergent sums; $a = \mathbf{1}_A$ and $b = \mathbf{1}_B$ are indicator functions of two subsets of $\mathbb{N}$, so $m_a, m_b$ are coordinate projections. Let $\mathcal{O} := \Phi(a, b; a, b) \in F$. Splitting $a = a^{\le t} + a^{>t}$, $b = b^{\le t} + b^{>t}$ and using multilinearity, $\|\mathcal{O}|_{I_t \times I_t} - G_t\| \le \varepsilon_t$. All entries are bounded by $C$. Expanding the determinant multilinearly in columns, every term containing a column of the perturbation is at most $\varepsilon_t C^{2t-1} < 1$, so the ultrametric inequality gives

$$|\det \mathcal{O}|_{I_t \times I_t}| = |\det G_t| \ge 1 \qquad \text{for every } t.$$

On the other hand, every element of $F$ is a uniform limit of matrices of finite rank, because each $W(x, y)$ has rank $\le 2$. So $\mathcal{O} \in F$ is within $\eta := (2C)^{-1}$ of a matrix $R$ of some finite rank $r$. Expanding $\det(R + (\mathcal{O} - R))$ on $I_t \times I_t$ in the same way, terms with more than $r$ columns from $R$ vanish, so $|\det \mathcal{O}|_{I_t \times I_t}| \le C^r \eta^{2t-r} < 1$ once $t > r$. This contradiction proves Theorem 2. □

## 7. An explicit field

The purpose of this section is to construct a field satisfying (K1) and (K2). Towards that end, let $k_0 = \mathbb{F}_2$ (or any field of characteristic 2). For $j \ge 0$ let $k_0((t^{1/2^j}))$ be the field of *formal* Laurent series $\sum_{n \ge N} a_n t^{n/2^j}$ ($N \in \mathbb{Z}$, $a_n \in k_0$) in the single variable $t^{1/2^j}$. These fields form an increasing chain, and we let

$$\mathcal{P} := \bigcup_{j \ge 0} k_0((t^{1/2^j})), \qquad |f| := 2^{-\operatorname{ord} f},$$

be the field of **2-power Puiseux series**. Here $\operatorname{ord} f$ is the minimum $q$ for which $\operatorname{coeff}_{t^q} f \ne 0$. We set $|0| = 0$ by definition. Let $k$ be the completion of $\mathcal{P}$ for this (ultrametric) absolute value. This is characteristic 2 since $k_0 \hookrightarrow \mathcal{P} \hookrightarrow k$.

**Proposition 14.** *$k$ is not spherically complete.*

*Proof.* We exhibit a sequence of balls as in (K2). Put

$$\gamma_m := 1 - 2^{-(m+1)}, \qquad c_n := \sum_{m=0}^{n} t^{\gamma_m} \in \mathcal{P}, \qquad r_n := 2^{-\gamma_{n+1}}.$$

*The balls have the form required in* (K2). Since $c_{n+1} - c_n = t^{\gamma_{n+1}}$ we have $|c_{n+1} - c_n| = 2^{-\gamma_{n+1}} = r_n$. Since the $\gamma_m$ are strictly increasing to 1, the $r_n$ are strictly decreasing, and $r_n > 2^{-1} = \frac{1}{2}$.

*The intersection is empty.* Suppose that $a \in k$ satisfies $|a - c_n| \le r_n$ for all $n$. As $\mathcal{P}$ is dense in $k$ there is $y \in \mathcal{P}$ with $|a - y| < \frac{1}{2}$, and then for every $n$

$$|y - c_n| \le \max\big(|y - a|, |a - c_n|\big) \le \max\big(\tfrac{1}{2}, r_n\big) = r_n. \tag{2}$$

Now $y \in k_0((t^{1/2^j}))$ for some $j$. The exponent $\gamma_j = (2^{j+1} - 1)/2^{j+1}$ has an odd numerator, therefore $\gamma_j$ is not a multiple of $1/2^j$ and so $\operatorname{coeff}_{t^{\gamma_j}} y = 0$. For $n \ge j$ we have $2^{-\gamma_j} > 2^{-\gamma_{n+1}} = r_n$. Furthermore, the term $t^{\gamma_j}$ occurs in $c_n$, so

$$\operatorname{coeff}_{t^{\gamma_j}}(y - c_n) = 0 - 1 \ne 0, \qquad \text{hence} \qquad \operatorname{ord}(y - c_n) \le \gamma_j \quad \text{and} \quad |y - c_n| \ge 2^{-\gamma_j} > r_n,$$

which contradicts (2). □

## 8. Remarks

*Remark* 15 (Why characteristic 2, why $\ell^\infty$, why this $F$). In characteristic $\neq 2$ the lift $\frac{1}{2}\big(m\circ(f,g)+m\circ(g,f)\big)$ exists, and the parity argument breaks (pairs add instead of cancelling). If $E$ has an orthogonal base $(e_i)$, e.g. $E=c_0$, and $F$ is Banach, then

$$P(f,g)(m)(x,y)=\sum_{i<j}(x_iy_j-x_jy_i)\,m(fe_i,ge_j)$$

is a bounded lift in every characteristic, so $Q$ is analytic; a similar formula works for a base of $E'$, and $t$-orthogonal bases suffice. This covers finite-dimensional $E$ and all Banach spaces over a discretely valued $k$ [7], so the counterexample needs a dense valuation; one can also show that $Q$ is analytic when $F$ is spherically complete. On $c_0$ the glued vectors $a,b$ of Step 4 do not exist and Proposition 13 fails. For $E=\ell^\infty$ and the larger target $\ell^\infty(\mathbb{N}\times\mathbb{N},k)\supseteq F$ a bounded lift also exists (sort the indices entrywise), but it sends $W$ to matrices such as the "staircase" $\mathbf{1}_{\{(p,q)\,:\,p\text{ odd},\,q\text{ even},\,p<q\}}$, which are not limits of finite-rank matrices, so it leaves $F$. These positive statements are not proved here and are not part of the formalization.

*Remark* 16 (What is forced). No single pair $(x,y)$ works for every lift: adding to $\Phi$ a bounded 4-linear map with values in $F$ that is alternating in $(a,b)$ and in $(x,y)$ gives another admissible $\Phi$, and this can change a given off-diagonal entry of $G(x,y)$ by an arbitrary scalar. Sets $I$ of bounded size cannot give a contradiction either: for $u=\mathbf{1}_{\text{odd}}$, $v=\mathbf{1}_{\text{even}}$ the rank-two matrix $W(u,v)\in F$ has the entry 1 at every position $(2s-1,2s)$. On a finite set $I$ the sorted lift of the previous remark has norm 1 and satisfies all the identities of Step 2, so there is no finite-dimensional obstruction: the invariant is the rank of one element of $F$ against the size of the sets $I_t$, and Step 3 is what allows infinitely many of them to be glued into one element.

*Remark* 17 (Higher degree, other characteristics). On paper, in characteristic 2 every $|\iota|\geq 2$ is covered by wedging with a coordinate of $E\oplus k$; this is not formalized. In odd characteristic $\ell$, for forms of degree $\ell$, the question is open: odd-degree forms have no Pfaffian-type top power, so a different invariant is needed.

## 9. Formalization

The Lean 4/Mathlib development is at

https://github.com/Deicyde/alternating-analytic-counterexample

The library `AltCounter/` contains no `sorry`, and the only axioms it uses are the three standard ones (`propext`, `Quot.sound` and `Classical.choice`). Its headline theorem assumes that Mathlib's

`ContinuousAlternatingMap.compContinuousLinearMapCLM`

is $C^\omega$ for all nontrivially normed fields, normed spaces and finite index types, and derives `False`; a second theorem does the same assuming this only for complete ultrametric fields of characteristic 2, Banach spaces, $E'=E$ and $|\iota|=2$. Both are restated in a Mathlib-only file `Challenge.lean` (with `sorry` in place of the proofs, as that format requires) and checked against `Solution.lean` by the `leanprover/comparator` tool. Steps 1–4 and §7 are formalized, including the complete proof of Proposition 13 by induction on $d$; Lemma 10 is `sum_eval_cubePt`, and the field of §7 is built with $k_0=\mathbb{F}_2$. The Lean constants in Step 4 are chosen with more slack than here, and the Lean form of Corollary 11 records only that the witnesses lie in the unit ball, which is all that Step 4 uses. Not formalized: non-analyticity at base points other than 0, index types with more than two elements, the positive statements in

the remarks, and the bundle-level consequence (no $C^\omega$ vector-bundle structure on bundles of continuous alternating maps in this generality).

## References


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